Phase 1: 6-V Lead-Acid Battery Charger

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***Abstract*—The following report presents the simulation stage for a 6-V lead-acid battery charger based on a discrete series-pass regulator with voltage feedback control. Before proceeding to physical construction, the circuit's functionality was verified through simulation in LTspice. The design relies on key components, including a Zener diode to provide a stable voltage reference, a transistor serving as an error amplifier for feedback comparison, and a pass transistor responsible for regulating the final output. To evaluate the circuit's stability, a DC sweep analysis was conducted by varying the input voltage from 10V to 30V. The results confirmed the design's effectiveness, showing a stable output maintained at approximately 7V. This successful simulation demonstrates that the feedback mechanism operates correctly to ensure reliable voltage regulation, providing a strong foundation for the future hardware implementation and testing phase.**

I. Introduction

Voltage regulation ensures a stable DC output despite variations in input voltage or load. Regulators are typically **linear** or **switching**. Linear regulators, such as series-pass and shunt types, offer low output noise and simple design, making them suitable for low-power and analog applications [1]. Switching regulators are more efficient but more complex and generate higher electrical noise [2].

A **series-pass regulator** maintains constant output by controlling a pass transistor in its active region. A Zener diode provides a reference voltage, while an error amplifier compares the output with this reference to adjust regulation [3]. Feedback loops improve accuracy and stability under varying conditions [4]. These regulators are commonly used in battery chargers and regulated power supplies [5].

This report details the design and simulation of a discrete series-pass voltage regulator for charging a 6-V lead-acid battery. The proposed circuit is composed mainly of a Zener diode to establish a stable voltage reference, a series-pass transistor to regulate both the charging voltage and current delivered to the battery, a driver base transistor, and a transistor-based error amplifier to compare the output against this reference . The design was verified through LTspice simulation, and the regulator's performance was evaluated under varying input voltages and load conditions before hardware implementation.

II. Circuit Design

The circuit design is about discrete series-pass voltage regulator for charging a 6-V lead-acid battery. For the circuit design, Table I shows the components used to construct the circuit. Figure 1 shows the circuit constructed using LTspice.

TABLE I
Components of the Circuit

Components	Use
DC Voltage source	To provide a sweep input from 10V to 30V
Capacitors	To keep the input voltage steady and stabilize the reference voltage and feedback voltage.
Zener Diode (Vz = 5.1V)	To maintain constant voltage.
Transistors	One is used as a series pass transistor, the second is used as a driver transistor, and the last is used for error amplification.
Resistors	To control the current across the circuit and the output voltage.

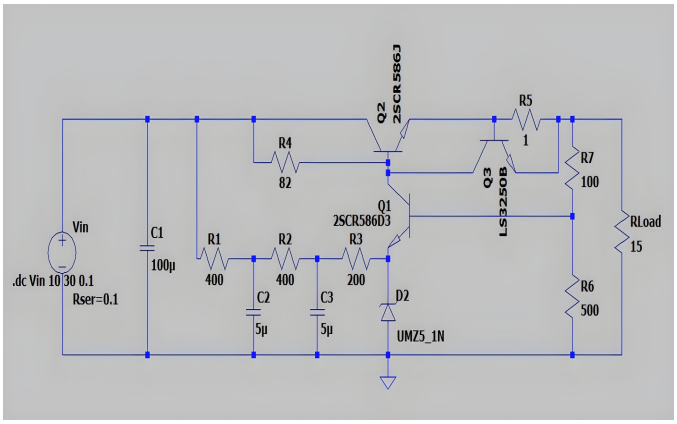


Fig. 1. A discrete series-pass voltage regulator.

The 100µF is connected in parallel with the voltage source to keep the input voltage steady, while the other 5µF capacitors are used to smooth the regulated output and prevent oscillations. Moreover, the Zener diode is used to maintain a constant 5.1 V reference voltage, which is then compared to a portion of the output voltage taken through the feedback divider formed by R_6 and R_7. For the transistors, Q_1 is the series pass transistor that controls and regulates the flow of current from the input to the output to keep the voltage steady. Q_2's mission is to amplify the small control signal from Q_3 to provide enough base current for Q_1. Q_3 acts as the error amplifier, comparing the output voltage with the reference and adjusting Q_2's drive to maintain a constant output.

Furthermore, Q_3 and R_5 act as an overcurrent protection, where R_5 serves as a current-sensing resistor that creates a

small voltage when load current rises. This voltage is applied to Q_3, causing it to conduct more and reduce Q_2's base drive, which limits Q_1's conduction and protects the circuit from overload. Meanwhile, R_1–R_3 provide the correct bias voltages and currents for the Zener diode and the transistors, ensuring that each device operates correctly, where they make the Zener voltage stable by controlling the current and the base and emitter voltage of the transistors in their active region. In addition, R_4 is used as a current limiting resistor for Q_2 as it restricts the amount of current flowing into Q_2's base, protecting the transistor from excessive drive and ensuring it operates within proper base current so that no overflow of current happens.

By using these components and exploiting their functions, the circuit can maintain a constant output voltage even when the input voltage changes.

III. Results and Discussion

LTspice was used to design the discrete series pass regulator circuit, simulate it, and evaluate the voltage regulation output of the 6-V lead-acid battery charging experiment. The functionality of the circuit was tested through different scenarios when the input voltage sweeps from 10V to 30V.

The first test was to examine the output voltage when there is no load (open circuit) as shown in Figure 2, to ensure that the output voltage is approximately constant despite the varying input voltage.

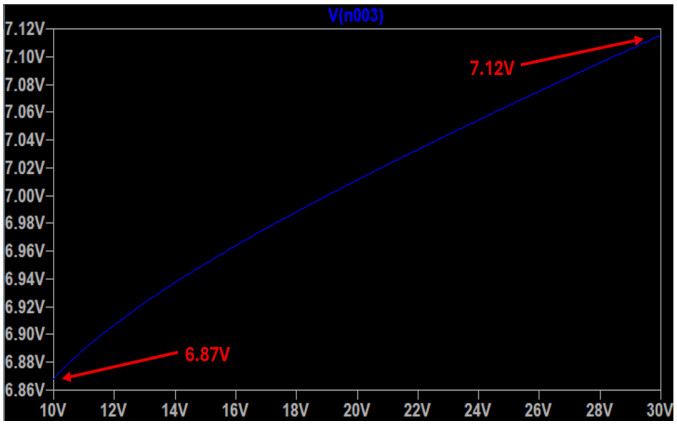


Fig. 2. The output voltage when there is no load (open circuit).

As observed in Table II , the value of output voltage is around 7V regardless of the change in the input value, which verifies the design functionality.

The second test conducted was to test the regulated voltage across different values of load resistors (5Ω, 10Ω, 15Ω, and 20Ω) and examine the voltage sweep across each resistor while the input voltage changes.

It is concluded from Table II that our regulator circuits work better with higher load values since the output voltage remained nearly constant. Figure 3 shows the output voltage for the 10Ω resistor. At a lighter load (5Ω), the output is observed to have a big sweep when varying the input voltage compared to the rest loads as illustrated in Figure 4. This is due to increased collector current through the pass transistor and power dissipation limits. This behavior confirms that the regulator performs best for moderate to high load conditions typical of a 6-V lead-acid battery.

TABLE II
Load VS Voltage and Current output

Load	Output Voltage (V)	Output Current (A)
Open Circuit	6.87-7.12	0
5Ω Resistor	5.2-7	1.03–1.4
10Ω Resistor	6.85–7.1	0.685–0.710
15Ω Resistor	6.85-7.1	0.457–0.474
20Ω Resistor	6.85-7.1	0.343–0.355

We also calculated the output resistance and see the difference between the original and the measured resistance as shown in Table III.

TABLE III
The output resistance

Output Voltage (V)	Output Current (A)	R	Error (%)
5.2-7	1.03–1.4	4.86	2.8
6.85–7.1	0.685–0.710	10	0
6.85-7.1	0.457–0.474	14.7	2
6.85-7.1	0.343–0.355	20.8	4

Overall, the observed results show that there is a small variance between the original resistance and the output resistance, indicating that the circuit maintained a steady output voltage and current across different resistance values.

Moreover, the voltage regulation percentage (VR) can be estimated as:

$$\%VR = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100 \tag{1}$$

which for the 10Ω load gives:

$$\%VR = \frac{7.1 - 6.84}{6.84} \times 100 \approx 3.8\%. \tag{2}$$

This indicates satisfactory regulation for a discrete analog design.

The third test used was to examine the circuit when there is a voltage source and a resistor in series to simulate a rechargeable battery as shown in Figure 5. Two different voltage values for the voltage source were used along with a 2Ω resistor to compare the current running through them and check if the circuit is functioning (5.7V to 7.12V).

It is observed that the current running for a lower voltage is considerably higher compared to a higher voltage battery as shown in Figure 6. This indicates that a lower voltage needs more current to be charged, unlike the charged battery, which will need less current as illustrated in Figure 7.

To evaluate how effectively input power is converted into useful output power, we will calculate the efficiency using:

$$\eta = \frac{P_o}{P_{in}} \times 100$$

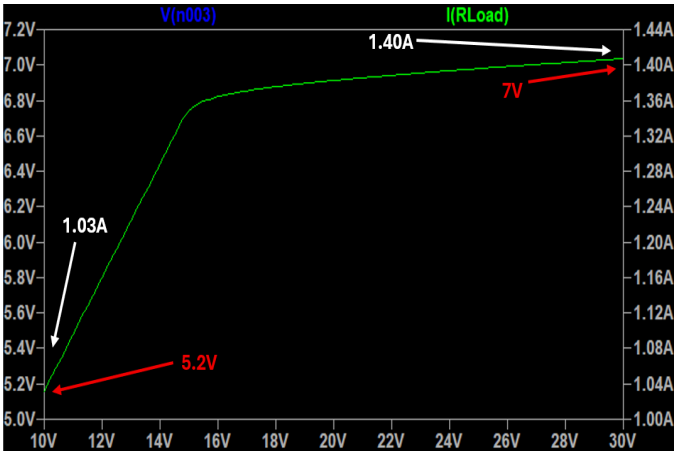


Fig. 3. The output voltage and current for the 5Ω resistor.

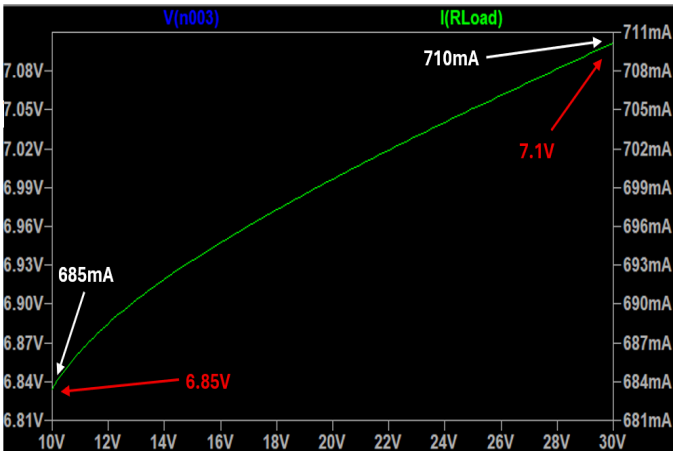


Fig. 4. The output voltage and current for the 10Ω resistor.

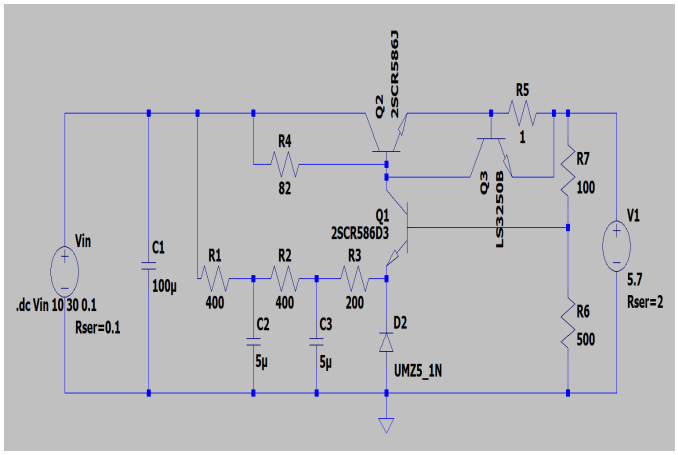


Fig. 5. The circuit when replacing the load resistance with a voltage source in series with a 2Ω resistor.

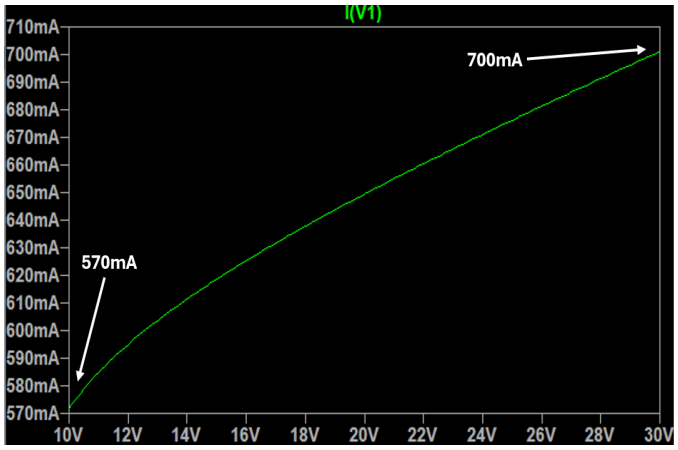


Fig. 6. The output current when the voltage is set to 5.7V.

where

$$P_o = V_o I_o \quad \text{and} \quad P_{in} = V_{in} I_{in}$$

The point that we will take to evaluate the efficiency is when R_L = 10Ω and V_{in} = 12V. Substituting the measured values of voltage and current:

$$V_{in} = 12\,V, \quad I_{in} = 750\,\text{mA} = 0.75\,A$$

$$V_o = 6.885\,V, \quad I_o = 688.5\,\text{mA} = 0.6885\,A$$

$$P_o = 6.885 \times 0.6885 = 4.74\,W$$

$$P_{in} = 12 \times 0.75 = 9\,W$$

$$\eta = \frac{4.74}{9} \times 100 = 52.67\%$$

However, efficiency isn't the most important factor in a linear regulator; this is because the series pass transistor continuously dissipates the voltage difference between input and output as heat while maintaining approximately a constant output voltage.

The feedback network effectively stabilized the output voltage by comparing the sampled output with the Zener reference through the error amplifier transistor. The pass transistor operated in the linear region, adjusting its base current to maintain the output voltage around 7V despite input conditions between 10V and 30V. At higher loads (15Ω and 20Ω), the voltage regulation was nearly ideal, with a small deviation (≈ 0.3V). However, at higher current loads (5Ω), the output voltage sweep was considerably higher due to the increased voltage drop across the pass transistor and limited drive current from the error amplifier stage.

IV. Conclusion

The 6-V lead-acid battery charger worked successfully using the discrete series pass regulator with voltage-feedback control. The circuit was checked and analyzed in LTspice using the DC sweep simulation, where the input varied between 10V and 30V to check the performance and regulation of the circuit, as well as how it behaves under different supply and load

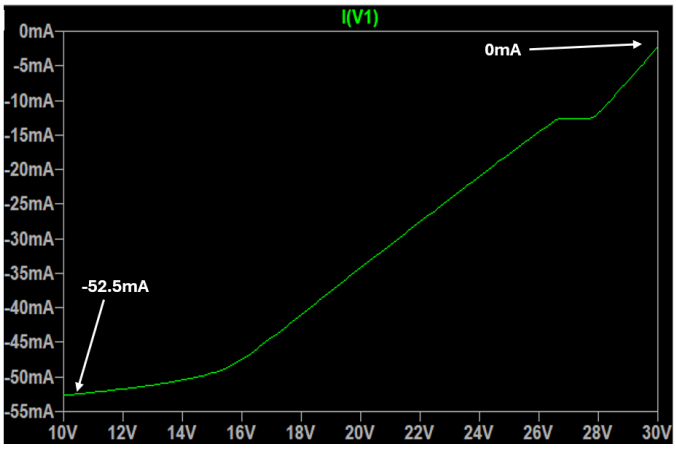


Fig. 7. The output current when the voltage is set to 7.12V.

conditions. Through the simulation results, we confirmed that the regulator maintained a stable output voltage of approximately 7V across a wide range of input voltages, proving the effectiveness of the feedback loop and zener reference in maintaining voltage stability. Furthermore, the output voltage had approximately a ±3-4% variation for higher resistance loads (10Ω-2x0Ω), showing amazing regulation suitable for the constant-voltage charging phase of a 6V lead acid battery. Due to transistor power limitations, minor voltage sag was observed under low resistance loads (5Ω), which is consistent with theoretical expectations for discrete linear regulators. And as a future work and improvement for the project, we will add two status LEDs - red for charging state, and green for the fully charged state.

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